

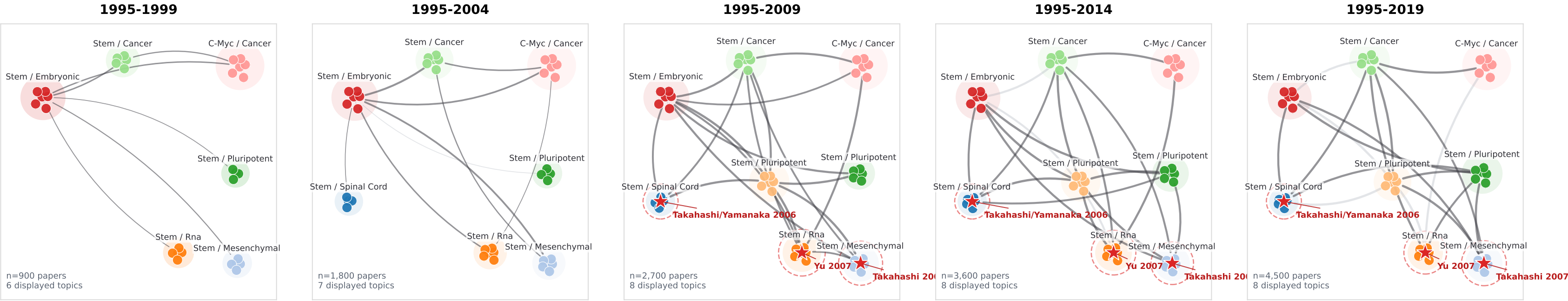
Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

Example domain: iPSC and cellular reprogramming. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots

iPSC and cellular reprogramming



Stem-cell and cloning priors

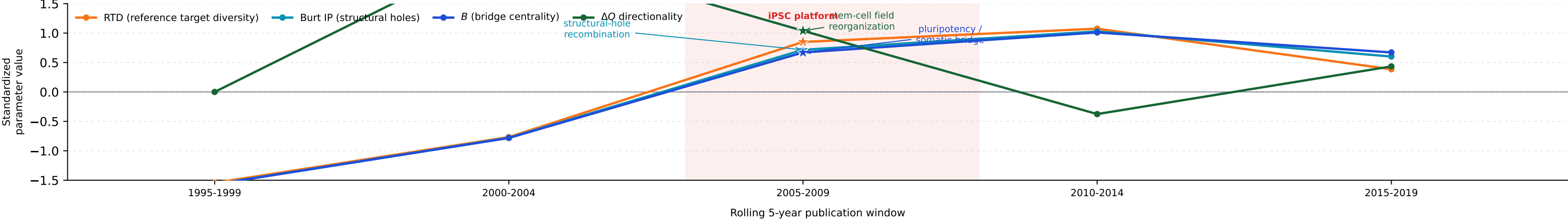
Factor-based reprogramming

Innovation shock: iPSC platform

Disease-model reconfiguration

Compression into regenerative hubs

b Dominant parameter trajectories explaining the graph transitions in panel a



Recombination
RTD up + Burt IP up
Structural holes are recombined

Bridging
B up
Somatic identity meets pluripotency

Reorganization
DeltaQ < 0
Stem-cell boundaries are reorganized